

FULL DSA MASTER PLAN (Product-Based MNC Ready)

PHASE 1 — Fundamentals (Arrays + Strings + Math)

Arrays: Traversal, insertion, deletion, searching, rotation, prefix sum, Kadane's algorithm, sliding window basics.

Strings: Palindrome, reverse, anagram, frequency array, two pointers, substrings.

Math: Prime numbers, GCD, fast exponentiation, modular arithmetic.

PHASE 2 — Linked List

Singly LL, reverse, find middle, cycle detection, merge sorted lists, add numbers.

PHASE 3 — Stack & Queue

Stack: Balanced parentheses, next greater element.

Queue: Circular queue, deque, sliding window max.

PHASE 4 — Hashing

Frequency map, two-sum, subarray sum, longest substring, group anagrams.

PHASE 5 — Recursion & Backtracking

Subsets, permutations, N-Queens, Sudoku solver.

PHASE 6 — Trees

Binary trees: Traversals, height, diameter, views, level order, build tree.

BST: Insert, delete, search, LCA.

PHASE 7 — Heaps

Min/max heap, k largest, k closest, merge k lists.

PHASE 8 — Graphs

BFS, DFS, cycles, bipartite, topological sort, Dijkstra, Bellman-Ford, MST, DSU.

PHASE 9 — Dynamic Programming

Knapsack, LIS, LCS, edit distance, matrix paths.

PHASE 10 — Patterns

Two pointers, sliding window, prefix sum, greedy, bit manipulation.

PHASE 11 — Practice

150–250 LeetCode problems, Striver SDE Sheet, mock interviews, contests.

ORDER:

Arrays → Strings → Math → Linked List → Stack → Queue → Hashing → Recursion → Backtracking
→ Trees → BST → Heaps → Graphs → DP → Patterns → Mocks.